

EC824A

Tests for Nonlinear Restrictions

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Nonlinear Testing

- We saw a generic nonlinear testing framework, $h(\cdot) \in \mathbb{R}^q$,

$$H_0 : h(\theta_0) = 0$$

vs.

$$H_1 : h(\theta_0) \neq 0$$

- For the Wald, LR, LM to be well defined, we need,

Assumption 1 (NR)

(i) $h(\theta_0)$ continuously differentiable on neighbourhood of θ_0 ; (ii) $H = \frac{\partial}{\partial \theta'} h(\theta_0)$ is full rank $q \leq p$; (iii) Ω_0 is nonsingular.

- For example,

$$h(\theta) = \theta_1 - \theta_2; \quad h(\theta) = \theta_1\theta_2 - \theta_3; \quad h(\theta) = \begin{pmatrix} \theta_1 \\ \theta_2 + \theta_3 - 1 \end{pmatrix}$$

Ω_0 nonsingular ensures asymptotic variance of $\sqrt{n}(\hat{\theta}_n - \theta_0)$ nonsingular

- The Wald statistic for this H_0 is defined, where $\hat{B}_n \xrightarrow{p} B_0$, $\hat{\Omega}_n \xrightarrow{p} \Omega_0$,

$$W_n = nh(\hat{\theta}_n)'(\hat{H}\hat{B}_n^{-1}\hat{\Omega}_n\hat{B}_n^{-1}\hat{H}')^{-1}h(\hat{\theta}_n) \quad \text{where } \hat{H} = \frac{\partial}{\partial \theta'}\hat{h}(\hat{\theta}_n)$$

- Intuitively, if H_0 is true, this quadratic form close to 0 (i.e. $h(\hat{\theta}_n) \in B_\epsilon(0)$)
- As usual, large sample theory gives us critical values for when this is sufficiently large

LM and LR

- LM and LR statistics use *restricted* extremum estimator:

$$\tilde{\theta}_n \in \Theta, \quad h(\tilde{\theta}_n) = 0, \quad \text{and}, \quad Q_n(\tilde{\theta}_n) \leq \inf\{Q_n(\theta) : \theta \in \Theta, h(\theta) = 0\} + o_p(n^{-1/2})$$

- LR test (in log-likelihood, and criterion function, respectively)

$$LR = 2(\ell(\tilde{\theta}_n) - \ell(\hat{\theta}_n)), \quad LR = 2n(Q_n(\tilde{\theta}_n) - Q_n(\hat{\theta}_n))/\hat{c}_n$$

we define $\hat{c}_n$ later (e.g. the error variance in linear model ML)

- LM test

$$LM = n \frac{\partial}{\partial \theta'} Q_n(\tilde{\theta}_n) \tilde{B}_n^{-1} \tilde{H}' (\tilde{H} \tilde{B}_n^{-1} \tilde{\Omega}_n \tilde{B}_n^{-1} \tilde{H}')^{-1} \tilde{H} \tilde{B}_n^{-1} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n), \quad \text{where} \quad \tilde{H} = \frac{\partial}{\partial \theta'} h(\tilde{\theta}_n)$$

where $\tilde{B}_n$ and $\tilde{\Omega}_n$ are consistent estimators of B_0 and Ω_0 under H_0

Tests for Nonlinear Restrictions

- In this lecture we study the asymptotic behaviour of the tests with critical values/rejection regions in mind
- Last lecture studied asymptotic behaviour of tests under null and local alternative to gauge “power”, or “consistency” of tests
- Here we are interested in asymptotic behaviour under the null, to generate critical values that are asymptotically valid

1 Nonlinear Testing

Nonlinear Testing – Assumptions

- Assumption 1 from earlier (continuously differentiable $h()$, $h'()$ full rank, Ω_0 nonsingular)

Assumption 2 (LR)

(i) $\Omega_0 = cB_0$ scalar constant $c \neq 0$; (ii) $\hat{c}_n \xrightarrow{P} c$ for non-zero random variables $\{\hat{c}_n : n \geq 1\}$

- E.g. Correctly specified ML, information matrix equality: $\Omega_0 = B_0$; $\hat{c}_n = c = 1$
- LS cond. homosked. ($E[U_i|X_i] = 0$, $E[U_i^2|X_i] = \sigma^2$), $c = \sigma^2$,

$$\hat{c}_n = \frac{1}{n} \sum_{i=1}^n (Y_i - g(X_i, \hat{\theta}_n))^2$$

- GMM, MD, TS with optimal weights with $\hat{c}_n = c = 1$ since $\Omega_0 = B_0 = \Gamma_0' V_0^{-1} \Gamma_0$

Nonlinear Testing – Assumptions

$$LR = 2n(Q_n(\tilde{\theta}_n) - Q_n(\hat{\theta}_n))/\hat{c}_n$$

- Intuitively, when H_0 true: $\tilde{\theta}_n$ close to $\hat{\theta}_n$, and continuity of Q_n means $Q_n(\tilde{\theta}_n) - Q_n(\hat{\theta}_n)$ close to zero
- Under H_1 , LR should be large since $\tilde{\theta}_n$ minimises Q_n over the restricted space, which does not include θ_0 ; hence, LR test rejects the H_0 when statistic adequately large.
- Sample equivalent of Assumption 2: $\tilde{\Omega}_0 = \hat{c}_n \tilde{B}_n$ and $\frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) = \tilde{H}' \tilde{\lambda}_n$ for;

$$\tilde{\theta}_n = \underset{\theta \in \Theta, h(\theta)=0}{\operatorname{argmin}} Q_n(\theta) \quad \text{vis a vis} \quad \mathcal{L}(\theta) = Q_n(\theta) - h(\theta)' \lambda$$

$$LM = n \frac{\partial}{\partial \theta'} Q_n(\tilde{\theta}_n) \tilde{B}_n^{-1} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) / \hat{c}_n = n \frac{\partial}{\partial \theta'} Q_n(\tilde{\theta}_n) \tilde{\Omega}_n^{-1} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n)$$

Nonlinear Testing – Assumptions

- For LR and LM we also need consistency of $\tilde{\theta}_n$ under H_0 “Restricted EE”

Assumption 3 (REE)

$\tilde{\theta}_n \xrightarrow{p} \theta_0$ under the null, H_0

- Consistency can be verified from first lecture on consistency, restricting the parameter space to $\tilde{\Theta} = \{\theta \in \Theta : h(\theta) = 0\}$
- Under H_0 , $h(\theta_0) = 0$ holds, hence $\Theta_0 \cap \tilde{\Theta} \neq \emptyset$ (plus regularity in $h(\theta)$)
- For W and LM also need consistency of Ω_0 and B_0 estimators under H_0

Assumption 4 (COV)

For W , $\hat{B}_n \xrightarrow{p} B_0$, and $\hat{\Omega}_n \xrightarrow{p} \Omega_0$ under H_0

For LM $\tilde{B}_n \xrightarrow{p} B_0$, and $\tilde{\Omega}_n \xrightarrow{p} \Omega_0$ under H_0

Nonlinear Testing – Critical Values

Theorem 1

(i). Under EE2, CF, NR, and COV, q restrictions,

$$W \xrightarrow{d} \chi_q^2 \text{ under } H_0.$$

(ii). Under EE2, CF, NR, COV, and REE,

$$LM \xrightarrow{d} \chi_q^2 \text{ under } H_0,$$

(iii). Under EE2, CF, NR, COV, REE, and LR,

$$LR \xrightarrow{d} \chi_q^2 \text{ under } H_0,$$

Nonlinear Testing – Critical Values

- For α level of significance (allowable Type I error) denote k_α^q as the $(1 - \alpha)$ th quantile for the χ_q^2 distribution

$$k_\alpha^q := \inf_{k \in \mathbb{R}_+} \{k : Pr(\chi_q^2 \geq k) \leq \alpha\}$$

- For each test, we can reject the null if the test statistic is greater than k_α^q
- Asymptotic theory in Theorem 1 says the sample test statistic will be rejected roughly $\alpha\%$ under the null: i.e. if the experiment under the null is repeated, close to $\alpha\%$ are rejected, when sample size is large
- Thought experiment: take a simulation of hypothesis tests when H_0 is true, with MC number of Monte Carlo rounds, and N sample size across each round of simulation. When N is large, we expect $\alpha \cdot MC$ number of simulated rounds to reject the null.

Nonlinear Testing – Critical Values

- Under local alternatives to H_0 , i.e. $H_1 : h(\theta_0) = C/\sqrt{n}$ with non-zero C
- W , LR , and LM are non-central Chi-squared asymptotically distributed with non-centrality metric

$$\delta_0 = h(\theta_0)'(HB_0^{-1}\Omega_0B_0^{-1}H')^{-1}h(\theta_0)$$

- This implies all three tests have the same large sample power functions
- Hence “in theory” use whichever is easier to compute!
- In finite samples (real world) tests can have different properties: Monte Carlo evidence suggests Wald over rejects (i.e. reject true null with above $\alpha\%$ rate); use LM and LR

Nonlinear Testing – Critical Values Wald

- Now to prove Theorem 1. Start with part (i): W

$$W = nh(\hat{\theta}_n)'(\hat{H}\hat{B}_n^{-1}\hat{\Omega}_n\hat{B}_n^{-1}\hat{H}')^{-1}h(\hat{\theta}_n)$$

- Taylor expand! θ^* is in between $\hat{\theta}_n$ and θ_0 , hence $\theta^* \xrightarrow{P} \theta_0$ follows from $\hat{\theta}_n \xrightarrow{P} \theta_0$

$$\begin{aligned}\sqrt{n}h(\hat{\theta}_n) &= \sqrt{n}h(\theta_0) + \frac{\partial}{\partial \theta'}h(\theta^*)\sqrt{n}(\hat{\theta}_n - \theta_0) = (H + o_p(1))\sqrt{n}(\hat{\theta}_n - \theta_0) + o_p(1) \\ &\xrightarrow{P} Z \sim N(0, HB_0^{-1}\Omega_0B_0^{-1}H')\end{aligned}$$

from $h(\theta_0) = 0$, and $\frac{\partial}{\partial \theta'}h(\theta^*) \xrightarrow{P} \frac{\partial}{\partial \theta'}h(\theta_0) = H$ under the null by Assumption NR.

- Convergence in distribution of $\sqrt{n}(\hat{\theta}_n - \theta_0)$ and continuous mapping theorem gives result
- Assumptions COV, REE, and NR: $\hat{H}\hat{B}_n^{-1}\hat{\Omega}_n\hat{B}_n^{-1}\hat{H}' \xrightarrow{P} HB_0^{-1}\Omega_0B_0^{-1}H'$
- Hence by Slutsky's Theorem, $W \xrightarrow{P} Z'(HB_0^{-1}\Omega_0B_0^{-1}H')^{-1}Z \sim \chi_q^2(0)$

Nonlinear Testing – Critical Values LM

- Taylor expansion + mean value theorem for $\frac{\partial}{\partial \theta'} Q_n(\tilde{\theta}_n)$ and $h(\tilde{\theta}_n)$, $(\tilde{\theta}_n, \tilde{\theta}_n^* \xrightarrow{P} \theta_0$ under H_0)

$$\sqrt{n} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) = \sqrt{n} \frac{\partial}{\partial \theta} Q_n(\theta_0) + \frac{\partial^2}{\partial \theta \partial \theta'} Q_n(\theta_0) \sqrt{n}(\tilde{\theta}_n - \theta_0)$$

$$0 = \sqrt{n} h(\tilde{\theta}_n) = \sqrt{n} h(\theta_0) + \frac{\partial}{\partial \theta'} h(\tilde{\theta}_n^*) \sqrt{n}(\tilde{\theta}_n - \theta_0) = \frac{\partial}{\partial \theta'} h(\tilde{\theta}_n^*) \sqrt{n}(\tilde{\theta}_n - \theta_0)$$

- $B_n(\theta) = \frac{\partial^2}{\partial \theta \partial \theta'} Q_n(\theta)$ and $H(\theta) = \frac{\partial}{\partial \theta'} h(\theta)$:

$$\begin{aligned} H(\tilde{\theta}_n^*) B_n(\theta^*)^{-1} \sqrt{n} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) &= H(\tilde{\theta}_n^*) B_n(\theta^*)^{-1} \sqrt{n} \frac{\partial}{\partial \theta} Q_n(\theta_0) + H(\tilde{\theta}_n^*) \sqrt{n}(\tilde{\theta}_n - \theta_0) \\ &= H(\tilde{\theta}_n^*) B_n(\theta^*)^{-1} \sqrt{n} \frac{\partial}{\partial \theta} Q_n(\theta_0) \xrightarrow{P} Z \sim N(0, H B_0^{-1} \Omega_0 B_0^{-1} H') \end{aligned}$$

by Assumptions CF(iii), CF(iv), REE, and NR and Slutsky's Theorem

Nonlinear Testing – Critical Values LM

- Assumption COV gives $\tilde{B}_n \xrightarrow{p} B_0$, $\tilde{\Omega}_n \xrightarrow{p} \Omega_0$
- Assumption REE gives $\tilde{\theta}_n \xrightarrow{p} \theta_0$
- Assumption NR gives differentiability (for Taylor expansions), plus full rank H , and invertible Ω_0
- If $\sqrt{n} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) = O_p(1)$, using $\tilde{H} \xrightarrow{p} H$, $\tilde{B}_n \xrightarrow{p} B_0$, it follows,

$$\tilde{H} \tilde{B}_n^{-1} \sqrt{n} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) \xrightarrow{p} Z \sim N(0, H B_0^{-1} \Omega_0 B_0^{-1} H')$$

- Conclude, for $Z \sim N(0, \Omega)$ and $\Omega = H B_0^{-1} \Omega_0 B_0^{-1} H'$

$$LM \xrightarrow{p} Z' \Omega^{-1} Z \sim \chi_q^2(0)$$

- This used $LM = \sqrt{n} \frac{\partial}{\partial \theta'} Q_n(\tilde{\theta}_n) (\tilde{A}^{-1} \tilde{A})' \tilde{\Omega}_n^{-1} (\tilde{A}^{-1} \tilde{A}) \sqrt{n} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n)$ where $\tilde{A} := \tilde{H} \tilde{B}_n^{-1}$

Nonlinear Testing – Critical Values LM

- Needed to show $\sqrt{n} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) = O_p(1)$; (shown $H(\tilde{\theta}_n^*) B_n(\tilde{\theta}_n^*)^{-1} \sqrt{n} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) = O_p(1)$)
- Lagrangian FOCs state (consistency implies *wpa1*, $\tilde{\theta}_n$ in interior of Θ):

$$\mathcal{L}(\theta) = Q_n(\theta) - h(\theta)' \lambda, \quad \Rightarrow \quad \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) - \tilde{H}' \tilde{\lambda}_n = o_p(n^{-1/2})$$

- This gives $H(\tilde{\theta}_n^*) B_n(\tilde{\theta}_n^*)^{-1} H(\tilde{\theta}_n)' \sqrt{n} \tilde{\lambda}_n = H(\tilde{\theta}_n^*) B_n(\tilde{\theta}_n^*)^{-1} \sqrt{n} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) + o_p(1) = O_p(1)$
- $H(\tilde{\theta}_n^*) B_n(\tilde{\theta}_n^*)^{-1} H(\tilde{\theta}_n)' \xrightarrow{p} H B_0^{-1} H'$, is nonsingular, hence $\sqrt{n} \tilde{\lambda}_n = O_p(1)$
- Thus, $\sqrt{n} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) = \tilde{H}' \sqrt{n} \tilde{\lambda}_n + o_p(1) = O_p(1)$ since $\tilde{H} \xrightarrow{p} H$ is full rank

Nonlinear Testing – Critical Values LR

- Expand $Q_n(\tilde{\theta}_n)$ around $\hat{\theta}_n$, with θ_n^* between $\tilde{\theta}_n$ and $\hat{\theta}_n$

$$\begin{aligned} LR = \frac{2n}{\hat{c}_n} (Q_n(\tilde{\theta}_n) - Q_n(\hat{\theta}_n)) &= \frac{2n}{\hat{c}_n} (\tilde{\theta}_n - \hat{\theta}_n)' \frac{\partial}{\partial \theta} Q_n(\hat{\theta}_n) + \frac{2n}{\hat{c}_n} (\tilde{\theta}_n - \hat{\theta}_n)' \frac{\partial^2}{\partial \theta \partial \theta'} Q_n(\theta_n^*) (\tilde{\theta}_n - \hat{\theta}_n) \\ &= o_p(1) + \frac{2n}{\hat{c}_n} (\tilde{\theta}_n - \hat{\theta}_n)' \frac{\partial^2}{\partial \theta \partial \theta'} Q_n(\theta_n^*) (\tilde{\theta}_n - \hat{\theta}_n) \end{aligned}$$

- This follows from $\sqrt{n} \frac{\partial}{\partial \theta} Q_n(\hat{\theta}_n) = o_p(1)$; and if $\sqrt{n}(\tilde{\theta}_n - \hat{\theta}_n) = O_p(1)$ is true.
- For $\sqrt{n}(\tilde{\theta}_n - \hat{\theta}_n) = O_p(1)$: result on last slide $\sqrt{n} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) = O_p(1)$

$$\Rightarrow \sqrt{n} \frac{\partial}{\partial \theta} Q_n(\hat{\theta}_n) + \frac{\partial^2}{\partial \theta \partial \theta'} Q_n(\theta_n^*) \sqrt{n}(\tilde{\theta}_n - \hat{\theta}_n) = O_p(1)$$

$$\Rightarrow o_p(1) + (B_0 + o_p(1)) \sqrt{n}(\tilde{\theta}_n - \hat{\theta}_n) = O_p(1)$$

since $\theta_n^* \xrightarrow{p} \theta_0$. B_0 nonsingular, hence $\sqrt{n}(\tilde{\theta}_n - \hat{\theta}_n) = B_0^{-1} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) + o_p(1) = O_p(1)$.

Nonlinear Testing – Critical Values LR

- Combining the following:

$$LR = o_p(1) + \frac{2n}{\hat{c}_n}(\tilde{\theta}_n - \hat{\theta}_n)' \frac{\partial^2}{\partial \theta \partial \theta'} Q_n(\theta_n^*)(\tilde{\theta}_n - \hat{\theta}_n); \quad \sqrt{n}(\tilde{\theta}_n - \hat{\theta}_n) = B_0^{-1} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) + o_p(1)$$

to show,

$$LR = \frac{n}{\hat{c}_n} \frac{\partial}{\partial \theta'} Q_n(\tilde{\theta}_n) B_0^{-1} B(\tilde{\theta}_n) B_0^{-1} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) + o_p(1) = \frac{n}{\hat{c}_n} \frac{\partial}{\partial \theta'} Q_n(\tilde{\theta}_n) \tilde{B}^{-1} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) + o_p(1)$$

which simplifies to $LR = LM + o_p(1)$

- Hence, the conclusion follows from LM result

Nonlinear Testing – LM

- The simplified LM can be derived to show,

$$\mathcal{L}(\theta) = Q_n(\theta) - h(\theta)' \lambda \quad \Rightarrow \quad \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) = \frac{\partial}{\partial \theta} h(\tilde{\theta}_n) \tilde{\lambda}_n = \tilde{H}' \tilde{\lambda}_n$$

Use from earlier, $H(\tilde{\theta}_n^*) B_n(\theta_n^*)^{-1} \sqrt{n} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) \xrightarrow{d} N(0, H B_0^{-1} \Omega_0 B_0^{-1} H')$, such that:

$$\begin{aligned} \sqrt{n} \tilde{\lambda}_n &= (H(\tilde{\theta}_n^*) B_n(\theta_n^*)^{-1} \tilde{H}')^{-1} H(\tilde{\theta}_n^*) B_n(\theta_n^*)^{-1} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) \\ &\xrightarrow{d} N(0, (H B_0^{-1} H')^{-1} H B_0^{-1} \Omega_0 B_0^{-1} H' (H B_0^{-1} H')^{-1}) \sim N(0, c(H B_0^{-1} H')^{-1}) \\ \Rightarrow \quad n \tilde{\lambda}_n' \tilde{H} \tilde{B}_n^{-1} \tilde{H}' \tilde{\lambda}_n / \hat{c}_n &\xrightarrow{d} \chi_q^2(0) \end{aligned}$$

- Hence, with $\sqrt{n} H(\tilde{\theta}_n)' \tilde{\lambda}_n = \sqrt{n} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) + o_p(1)$,

$$n \frac{\partial}{\partial \theta'} Q_n(\tilde{\theta}_n) \tilde{B}_n^{-1} \frac{\partial}{\partial \theta} Q_n(\tilde{\theta}_n) / \hat{c}_n \xrightarrow{d} \chi_q^2(0)$$